DNSSEC

by

Steven Dormady

An Independent Study Project

Department of Computer Science

James Madison University

Department of Computer Science

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Abstract

TBD

Introduction - Add this to Abstract?

1.1 Motivation

DNS by itself is insecure and vulnerable to attack. With DNSSEC, these vulnerabilities are contained and mitigated.

My personal motivation for this study was a hand on security project, proctored by Dr. Aboutabl, who has become one of my personal favorite professors here at JMU. His Information Security class helped further my interest for security, and when he mentioned this study, I knew I had to seize the opportunity.

1.2 Problem Statement

The problem we are trying to solve here is a dive into DNS Security Extension, or DNSSEC, and how it changes DNS. Using a testbed network of virtual machines on a virtual network, I was able to use Bind9 to be able to set up the DNS environment, then convert it to use DNSSEC. In this study of DNSSEC, I will be looking at the algorithms and data structures used in DNSSEC, and how they are used to secure DNS. I will also be looking at the performance of DNSSEC, and how it compares to DNS.

1.3 Contributions

What did I contribute?

1.4 VM Research

I have never dealt directly with Virtual Machines before, and building a network of them was something I had never really thought about. However, the thought excited me and it seemed like a great opportunity to learn a new skill. Setting up the first one was a bit of a learning curve, but after setting up the first one to the specifications I wanted, I was able to clone the rest as needed.

The software I used to set up the testbed network for my DNSSEC program was VMWare Workstation. This software made it very easy to set up the virtual machines and network. Having a built in option to create a virtual network was very nice. I was also able to connect to the internet and download the necessary tools I needed, like Bind9, VMWare Tools, Wireshark, gcc and more. I also chose to use git for version control and being able to edit the code on multiple devices if needed.

1.5 VM Network

On VM Workstation, it is very easy and simple to set up a Virtual Network. Under the ”Edit” tab in the menu, I was able to create a new vmnet and customize it to my needs. I made a custom network, isolated from the outside networks. The IP I used was 192.168.107.X, with a subnet mask of 255.255.255.0. The 192.168 is an IP reserved for local networks, so this was perfect for me to to use. There was an option to enable or disable DHCP service. Enabling it meant that you could let DCHP assign addresses to machines on the network. For me, I wanted to statically and manually assign IPs without and DHCP intervention, so I deselected this option to have more control in my environment.

1.6 Bind9

I installed Bind9 on all of the VMs in the Virtual Network. The version I installed was 9.18.39 (check). I decided on Bind9 because it was a good option to be used on Ubuntu machines and gave me the control I wanted.

Introduction to the Domain Naming System

2.1 DNS Background

Firstly, I think a bit of background about DNS helps to understand why it became what it is today. DNS started as ARPANET, which was very inefficient and used a centralized hosts file to map the domain names to IP addresses. When a change was made, it had to be redistributed to all of the systems. With number of users and systems needed increasing, this became very hard to maintain, which led to DNS today.[1]

In my research about DNS, I learned a lot about what happens behind the scenes with DNS Servers. Cloudflare called it, ”DNS is the phone book of the internet.”, and I think this name is very fitting.[2] Like a phonebook, you look up a name (domain name) and get someone’s phone number (IP address). All network systems operate with network addresses, such as IPv4 and IPv6. More or less, DNS translates domain names, like www.example.com, to IP addresses, like 134.126.126.99 (jmu.edu), so browsers can load internet resources.

The DNS naming structure is organized hierarchically in a tree-like format. Starting at the root of the tree, there is the Root Domain, which is where real world DNS begins. It delegates authority to the Top-Level Domains, or TLDs, which delegate further to Second-Level Domains, or SLDs, and continue onward, spreading out like a tree. Below in figure 2.1 is a diagram demonstrating this hierarchy.[1]

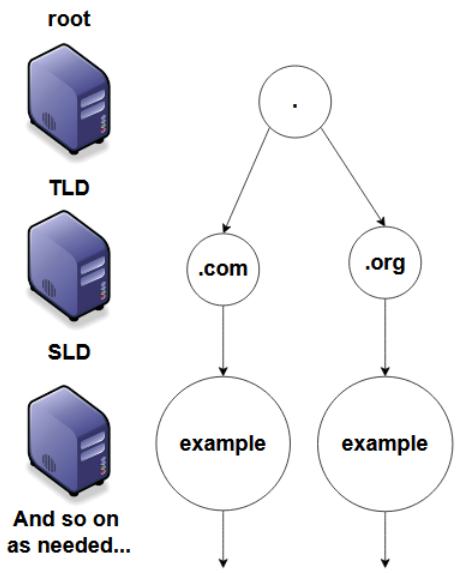

Figure 2.1: Root Delegation Hierarchy

A domain is the label of a node in the tree. A dot in a domain name means that it must traverse to another layer to get more information. Like in the above example, we start with ”example”, being at the Second-Level domain, then traverses to ”.com” at the Top-Level domain, then finally the root, ”.”, server. Put it all together and you get example.com., with the end period making the a Fully Qualified Domain Name, or FDQN. This period is usually hidden behind the scenes, and as users we do not use it. Root servers is the beginning of all DNS name resolutions, which makes the name root make sense. Like a tree, the DNS tree needs it’s roots to work. However, it helps to see the root domain being reached in this instance. There are 13 root servers due to historical reasons of how many addresses could be put into a 512 byte UDP message.[1]

2.1.1 DNS Resource Records

These DNS records are the most common records used in building a DNS server. These records are all in db files, which are database files used for each zone defined. For example, the zone ”.lab” will have a db.lab file. You can name it whatever, as long as it aligns with the named.conf.local zone path defined for the zone. It is best practice to have the domain

name as the db files extension. For both the A and NS records, their line begins with an @ symbol. Below is an example of my db.lab file for the lab zone on VM1.

```
steve@steve-VMware-Virtual-Platform:~$ cat /etc/bind/db.lab
;
; BIND data file for local loopback interface
;
$TTL    604800
@       IN      SOA     ns1.lab. root.lab. (
                              5         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      ns1.lab.

ns1     IN      A       192.168.107.129

jmu.lab.        IN      NS      ns1.jmu.lab.
ns1.jmu.lab.    IN      A       192.168.107.128
bc      IN      A       192.168.107.1
computer        IN      A       192.168.107.2
```

Figure 2.2: The db.lab file for the lab zone.

Start Of Authority Records

SOA records, or ”Start Of Authority” records, show the domain name that the specific db file has authority over. There are several sub records within. The MNAME is the domain name of the name server of the original or primary source of data for the zone. Like a .com, or for this study, .jmu and .lab. These are typically preceded with an ns1, which will come into play later with Namserver Records later. The next record is a RNAME, which is the Responsible Person record. For this project, I have root.jmu or root.lab for the root having responsibility. The next record is the serial number, which Bind uses to load different serial versions of the zone. This record has the space to be an unsigned 32 bit number. It should be updated every time there is a change to the db file. Some developers like to use the date it was edited on to help keep track of when it was last edited. Next, the refresh record is a 32 bit time interval before the zone should be refreshed. The next record is the retry record,

whish is another 32 bit time interval that would run out before a failed refresh should give up. Lastly, the expire record defines another 32 bit time value that gives an upper bound on the time interval that can run before the zone is no longer authoritative. Below is an example of an SOA record for BLANK zone.[3] An example cane be seen below in figure 2.3.

```
@       IN      SOA     ns1.lab. root.lab. (
                              5          ; Serial
                         604800          ; Refresh
                          86400          ; Retry
                        2419200          ; Expire
                         604800 )        ; Negative Cache TTL
```

Figure 2.3: An example of the SOA record in the parent servers db.lab, representing the start of authority for the zone.

Address Records

A records are address records, ”A” for short. These are the records you get when a domain maps to an IP address of a given domain. This is the most fundamental type of DNS record. There are also ”AAAA” records, which are for IPv6 traffic, but for this study we will only focus on IPv4 IPs. These records enable a user’s device to connect with and load a website, or whatever is at the IP address.[4] As seen below, in figure 2.4, when querying for an ip at one of my DNS Servers, I am returned with the IP address of that record.

```
steve@steve-VMware-Virtual-Platform:~$ dig @192.168.107.129 computer.lab A +short
192.168.107.2
```

Figure 2.4: An example of digging for an A record, the short flag used to only give me the ip.

Below, figure 2.5, is an example of what these records look like.

```
bc      IN      A       192.168.107.1
computer        IN      A       192.168.107.2
```

Figure 2.5: An example of A records in the parent servers db.lab, computer.lab and bc.lab

Nameserver Records

NS records stand for ”Nameserver” records, and indicate which DNS server is authoritative for that domain. Like in the SOA, ns1.lab shows which server is authoritative for the .lab domain. Domains can have multiple NS records, like .lab has, which indicate primary and secondary nameservers for domains. For instance, in db.lab, there are multiple NS records, one pointing the the server itself and another pointing to the secondary server for the .jmu.lab domain.[5] See figure 2.6 for an example.

```
@          IN      NS      ns1.lab.

ns1        IN      A       192.168.107.129

jmu.lab.           IN      NS      ns1.jmu.lab.
ns1.jmu.lab.       IN      A       192.168.107.128
```

Figure 2.6: An example of NS records in the parent servers db.lab

2.2 The DNS Environment

There are numerous similarities between DNS with the internet and this testbed study. The bridge between the user and DNS servers is the DNS resolver or name resolution. In the project, the client digs at a server, and gets a response. This is a higher level abstraction, as there are no websites mapped to the IPs on my DNS servers, just more local IPs which could be a website. There is also no official ”.” or root domain, but my ”root” domain is lab., which is also my top level domain. This is seen in figure 2.7. In this setup, authority is held in the parent vm and passed downward onto the child server. In real DNS, a recursive resolver is used and traverses the different levels of domains.

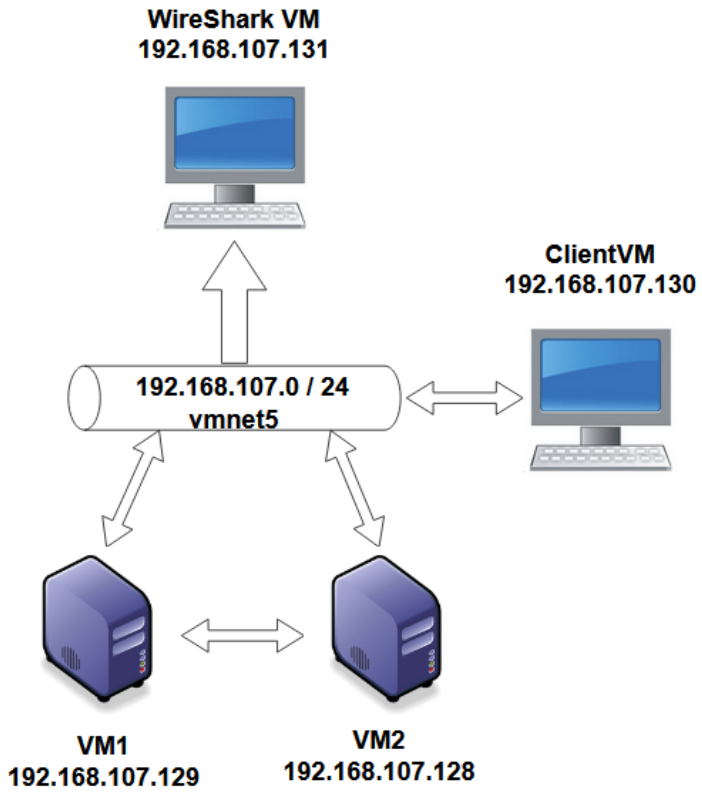

Figure 2.7: This is a diagram of the setup of the virtual environment, with inspiration of the layout coming from Dr.Aboutabl

2.2.1 Environment Zones

In the environment, there are different zones between the two name servers. As seen in figure 2.8, the parent zone holds .lab, with an extension to the child zone, .jmu. To be able to query the child, you need a domain name, followed by .jmu.lab. A full example of a valid query would be like this: quad.jmu.lab. Quad is the final piece, and this domain maps to 192.168.107.64. You can query the child server directly and receive responses, or you can query it through the parent server.

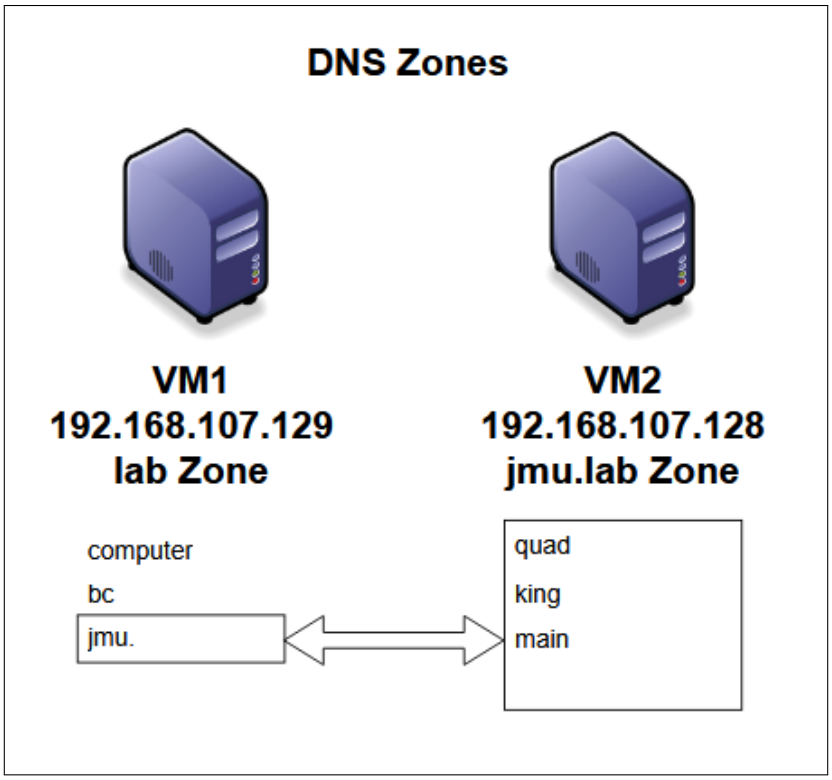

Figure 2.8: This is a diagram showing the zones and their relationship.

In this study, both the parent and the child server have authority over their own domains, and you can also traverse the tree to query the parent for information on the child server, as seen below in figure 2.9. There are also no IPV6 records, and this is disabled in the configuration of the servers.

```
;; AUTHORITY SECTION:
jmu.lab.                86400   IN      NS      ns.jmu.lab.

;; ADDITIONAL SECTION:
ns.jmu.lab.             86400   IN      A       192.168.107.128

;; Query time: 1 msec
;; SERVER: 192.168.107.129#53(192.168.107.129) (UDP)
;; WHEN: Tue Feb 24 22:57:16 EST 2026
;; MSG SIZE  rcvd: 102

steve@steve-VMware-Virtual-Platform:~$ dig @192.168.107.128 quad.jmu.lab A +short
192.168.107.64
```

Figure 2.9: Querying the parent for information about the child, the querying the child with that information.

2.3 DNS Vulnerabilities

DNS without DNSSEC is very insecure and vulnerable to a plethora of attacks. Without DNSSEC, queries are readable to any person on the internet. An attacker could easily intercept this traffic and replace it with malicious content. This is where DNSSEC comes into play, helping to mitigate and prevent many attacks from occurring. It also proves that the information is coming from a trusted, authoritative source.

2.3.1 DNS Spoofing/Cache Poisoning

This is an especially vulnerable part of DNS without DNSSEC. This attack is a textbook man-in-the-middle attack, when the attacked intercepts a query and replaces it with their malicious information. DNS spoofing is when an attacker inserts their malicious address in place of a cached address. This attack is using DNS Hijacking in this instance, to redirect to what the attacker wants them to see. They pretend to be a DNS nameserver and forge a reply, and with UDP it makes it even easier for this attack to occur. Cache poisoning is DNS Spoofing without having to hijack the query. The systems involved with DNS, like servers, computers, and routers can cache DNS lookups for quicker recall. An attacker can poison the cache by replacing an entry with a spoofed domain. This spoofed domain gets accessed until the cache gets refreshed and the spoofed site gets replaced.[6]
Acording to the article, cache poisoning is when the attacker injects bad data into a recursive nameservers cache, which leads to it distributing this data without knowing and the client does not know either. The host names are legit, but the server they are visiting cannot be trusted and is bad.

2.3.2 DNS Hijacking

DNS Hijacking occurs when an attacker tries to redirect where the client is going with their query. The can perform a man-in-the-middle attack and take the query and replace it with

their IP, causing the user's machine to directly communicate with the attacker. They could also give an IP that delivers a malicious payload. These are some of the most basic types of attacks when it comes the DNS hijacking.[7]

Introduction to DNS Security Extension

3.1 DNSSEC Background

DNSSEC, or Domain Name System Security Extension, adds a layer of security to the otherwise vulnerable DNS. DNSSEC adds security by using cryptographic signatures with public key cryptography, with the private key signing it and the public key verifying. DNSSEC also implements hashing, which helps to add more confusion and making it harder for attackers to know what plain text strings were previously used. It also adds data integrity and and authentication to the data, showing whether it was tampered with and the signing provides the authentication.

There is an additional flag with dnssec queries, the ”ad” flag. This flag is passed along with successful DNSSEC queries, meaning ”Authenticated Data”. When you query for a domain with dnssec enabled, receiving this flag back instead of a SERVFAIL status means that the data was successfully authenticated.

This is NOT True? Should revisit. Do experiment where it doesn't validate. It validates now, still need to make a copy key and corrupt it.

You will not see it in the below screen shots, and that is because there isn't any validation going on yet. Seeing the records is enough for this part of my project, where as the next chapter I will write code to validate these records.

Below, figure 3.1, is db.lab, but now set for DNSSEC. It changed directories to allow for

permission to be more for the owner of the server (me). Originally, DNS configuration and the db files were in /etc/bind/, but now they moved to /var/lib/bind/ to allow for more freedom and not having to deal with bind ownership. Changing the zone configuration to point to the new db file allows the server to write into this directory. As you can see below, there is a good number of changes to the db file. Gone are the @ symbols preceding lines, and there is a new minimum time record. There is still a SOA and NS and A records, which have minimally changed from regular DNS. At the bottom of the figure, yuo can see a Delegation Signer record, which came from the child zone. This is an important part of the Chain of trust, allowing the parent to verify the child is the child.[8]

```
$TTL    86400
lab.    IN      SOA     lab. root.lab. (
                              3         ; Serial
                          43200         ; Refresh
                            900         ; Retry
                        1814400         ; Expire
                           7200 )       ; minimum

lab.    IN      NS      ns.lab.
ns      IN      A       192.168.107.129

jmu     IN      NS      ns.jmu.lab.
ns.jmu  IN      A       192.168.107.128
; bc    IN      A       192.168.107.1
computer        IN      A       192.168.107.2
jmu IN DS 34939 8 2 9682051422A73107F882F98D04F47492ECA60648DC071428E43477AC7F1
42C2
```

Figure 3.1: The new DNSSEC db.lab

3.2 New DNS Resource Records

With the regular DNS records previously discussed, there are some new records that must be taken into account. Regarding the Zone Signing Key and the Key Signing Key, the algorithms I used for the keys was RSASHA256, this is a good algorithm for encryption and one I am familiar with. The ZSK signs the records from the zone, and the KSK signs the ZSK. The hand shake looks a bit like this, an example of a query for computer.lab, figure 3.2.

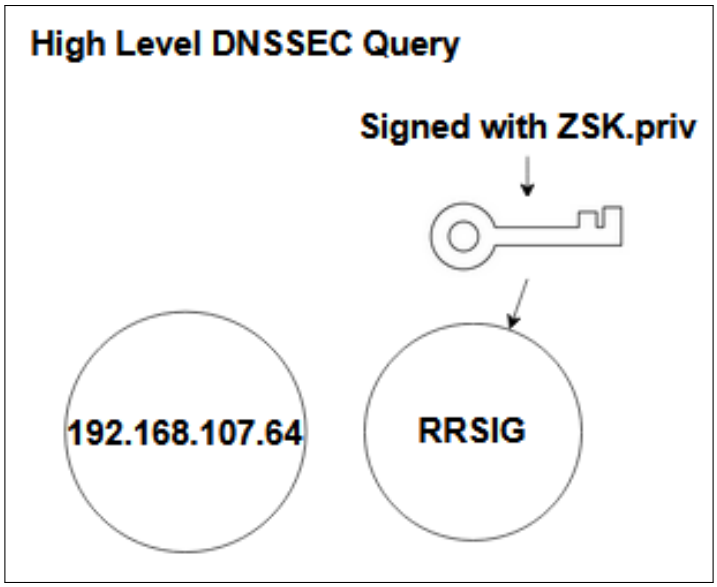

Figure 3.2: A diagram showing a high level view of a DNSSEC query, digging for an Address Record.

3.2.1 Resource Record Set

This record is not an explicit record, rather a descriptive word to describe records that are grouped together. For instance, if I query for my parent server for DNSKEYs, I receive the RRSet that is signed, seen below in figure 3.3. These keys are signed together as a set by the Key Signing Key, which the name is pretty explanatory for what it's purpose is. [9]

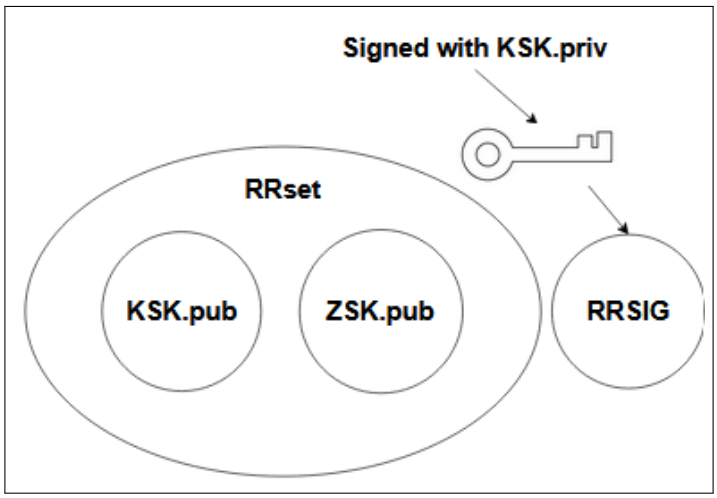

Figure 3.3: This query shows a dig at the parent for the DNSKEYs

3.2.2 Resource Record Signature

The Resource Record Signature, or RRSIG record, is used to verify the answers that are received. These are attached to every answer that is sent with DNSSEC. The signatures are used by recursive name servers to verify that the answer to the question is valid. [10] Below is an example of an address query with a signature, figure 3.4.

```
steve@steve-VMware-Virtual-Platform:~$ dig @192.168.107.129 computer.lab +dnssec +multiline

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> @192.168.107.129 computer.lab +dnssec +multiline
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 19718
;; flags: qr aa rd; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: do; udp: 1232
; COOKIE: 3df3cdc0dcb579af0100000069a5d721e88ca1e9eeb51c37 (good)
;; QUESTION SECTION:
;computer.lab.          IN A

;; ANSWER SECTION:
computer.lab.           86400 IN A 192.168.107.2
computer.lab.           86400 IN RRSIG A 8 2 86400 (
                                20260310233437 20260225023500 34104 lab.
                                GkosYckWQnxLuVJhdzymzDDRZuS+hBs5zzsdmrBCFkc3
                                4hdxuO8bbGjPQfkJTB9fXyov9xc6M1rDQko8+qzgD+9d
                                Kuextffp+oU3kiOLiJosuEUlDciTkVR/u4nm0T84STDN
                                RmK1yjPtp+hb4ZK0eLTIm3Ksqx/xQsT6Pl6Gl/Q= )

;; Query time: 0 msec
;; SERVER: 192.168.107.129#53(192.168.107.129) (UDP)
;; WHEN: Mon Mar 02 13:29:53 EST 2026
;; MSG SIZE  rcvd: 248
```

Figure 3.4: Address record with a RRSIG from the ZSK

The is a lot of information in the fields that come with an RRSIG resource record. After the computer.lab domain name and the TTL, 86400, the first field is the type, being a RRSIG for an address record. After that, there is the algorithm field, 8 standing for RSASHA256. The next field is the labels field, which represents the count of labels in the domain name, excluding the root. The next field is the original TTL of the RRset, which is the value it is given in the authoritative zone. The field that follows the TTL is the signature expiration and inception fields. These are 32-bit unsigned integers, represented in the number of seconds since January 1st, 1970, or like the format above, being YYYYMMDDhhmmss, years, months, days, hours in 24 hour format, minutes and seconds. The latter format is distinguishable by having exactly 14 digits. The seconds option with be a 32-bit integer

that won't be longer than 10 digits. The inception date specifies when it was created, and the expiration date being when it is no longer valid. Next is the key tag field, which holds the key tag of the signing key, in this instance the ZSK for the lab zone.

All of the metadata inside the RRSIG Record, from the type to the keytag, gets concatenated and then hashed, then signed with the ZSK. This creates the signature which is the last piece of the RRSIG resource record puzzle.

3.2.3 DNSKEYs In General

DNSKEYs are the record that are used to sign an validate RRsets. These keys use public key cryptography, storing the public keys in DNSKEY RRs. These keys are used in different ways, and there are two different keys that we care about in this study, which are the Zone Signing Key and the Key Signing Key. The identifying type number that this RR is a DNSKEY is 48.[9]

The general format for these RRs is as follows, figure 3.5

```
;; QUESTION SECTION:
;lab.                    IN DNSKEY

;; ANSWER SECTION:
lab.                    3600 IN DNSKEY 257 3 8 (
                                AwEAAdyubupVYO97+QR5LxX3eIh8vxOI5bxfVz6ieZqM
                                F19e5Ul2SD+MyyY5iM1VRutYEpLmfbMDDuW3ZElJzN4B
                                RP8huMuzRf8V4nAuB3poxIUs+kRywr4n6et46oCBF7oW
                                mcw792jAKKXFiuegwZ3s5NvrZufCckq5wzgaPvMxJLZL
                                kalddzkc0KuvwAtmGsOTT7lkn3sx59JgWGSYiL+3FNZF
                                SZPlcDyw3QuIO1hN+Vq2NwuSR2LkOFX3miCKx4d07hr2
                                7TtP3LGJiBhcyMeJRT3i4bOBbCtwGelA6xq5i4k1fg2+
                                R2e+2VZtqk35GJsh8c0yqKoahz96ZCSrWCHm690=
                                ) ; KSK; alg = RSASHA256 ; key id = 5852
lab.                    3600 IN DNSKEY 256 3 8 (
                                AwEAAbDODYwllXwQTD9OaDxw02d579560Yf4dGH7VpOb
                                +FcTTMIovHAAPuSgD0l5b1DnxWVhqupG9rM8b6FKxC+6
                                zNJycr9nUjTLkBxaVYgicGaS+X/0qy0K+d7spTg4qw3W
                                jzAls4LqquwljeMTNLe/peC2BZM/cmz0Mhn1MacYqqvH
                                ) ; ZSK; alg = RSASHA256 ; key id = 34104
lab.                    3600 IN RRSIG DNSKEY 8 1 3600 (
                                20260312172022 20260226164849 5852 lab.
                                sTOzJDy11sodWrqEb63MmZzOwhEfgO1ve7n/YgjL1ooQ
                                ECvhsNfYXM5D8Dq2R//oatu17DR9gT2fM591t/Hgi8lI
                                9B69/iPGOvArrKiK3PqmCRFjBnhl5T7ARcsvmA6Yedlj
                                yO2YitOs3aHYIZPoFhmRWrX0OwRA6rVO7/1JvdY/sZDI
                                Odi/oekocHkJ2pBdHP4tymB5nAiSn1Zm2RkGgNSJeQ1w
                                PvGccqXZIDuudTtI6Dy1+JTBZffvpYKwHupy3pCJJTdj
                                DNQ9rpsS15GlignoH0fX4B1WmhvHt5CHFWm5/tDyrXYh
                                XIsiOP9XSx6X8RojnKATV+3DilfgbMgXnQ== )

;; Query time: 0 msec
;; SERVER: 192.168.107.129#53(192.168.107.129) (UDP)
```

Figure 3.5: This query shows a dig at the parent for the DNSKEYs

These keys are stored in /var/cache/bind/, and are generated with the dnssec-keygen command. There are two parts of these keys, a ".key" and ".private", with the zone name and

key tag in the name. There is an example of these stored records below. The keys are generated with the algorithm specified in the dnssec policy, which is RSASHA256 in this case. Generally, the KSK is generated with a larger key size than the ZSK, because the KSK is used to sign the ZSK, which is used to sign the records. The KSK is more important to protect, because if it is compromised, then the ZSK can be compromised, and then all of the records can be compromised. The ZSK is used more frequently, so it is more likely to be compromised, but it is also easier to replace than the KSK.

Zone Signing Key

The Zone Signing Key, or ”ZSK” for short, is the key that is used to sign data from a zone. It signs all records except for RRset that are related to keys. The ZSK has the 256 flag, indicating that it is the ZSK. This key has a shorter Time To Live, or TTL, because it is sent out and used much more frequently than the Key Signing Key. Seen later under DNSSEC Policy, I set the time to live for the ZSK to be 60 days, where as the KSK is set to be a year. From what I saw, this is pretty standard.

Key Signing Key

The Key Signing Key, or ”KSK”, is used to sign what the ZSK does not sign: key-related RRsets. This keys serves another purpose, as it is the bridge between a parent and a child zone. The flag for the KSK is 257, indicating that this key is the KSK. Like the Zone Key, this key also follows the public key encryption scheme. This key pair is meant to protect the ZSK for the zone being queried.

3.2.4 Delegation Signer

The Delegation Signer Record, or DS record, is used to ”vouch” for another zone. The parent holds a copy of this in their zone, and when asked to query about the child zone, it can verify that the response comes from the child. This record is a hash that is stored of the

child zone's KSK, and the parent can use this hash to verify that the child zone is the child zone. This is an important part of the Chain of Trust, allowing the parent to verify the child is the child, ensuring that the responses they receive can be trusted.[8]

These records are generated with the dnssec-dsfromkey command, which takes in the child zone's KSK public key and generates the DS record that can be stored in the parent zone. The output from this command gets pasted directly into the parents zone file, and then the parent can use this to verify the child zone's KSK. Below is an example of a DS record and the command used to generate it, figure 3.6. After adding this delegation signer record to the parent zone, I had to run an rndc reload the zone.

```
steve@steve-VMware-Virtual-Platform:~$ dnssec-dsfromkey -2 /var/cache/bind/Kjmu.
lab.+008+34939.key
jmu.lab. IN DS 34939 8 2 9682051422A73107F882F98D04F47492ECA60648DC071428E43477A
C7F1342C2
```

```
bc      IN      A       192.168.107.1
computer        IN      A       192.168.107.2
jmu.lab. IN DS 34939 8 2 9682051422A73107F882F98D04F47492ECA60648DC071428E43477AC7F1342C2
```

Figure 3.6: The first image is the command used to generate the DS on the child server, then the output in the parent zone file.

3.3 Chain Of Trust

The Chain of Trust is how different levels of servers know they can trust each other. There must be trust through the whole process of the query, otherwise there is a vulnerable spot for intrusion by an attacker.The trust starts with a fitting name, the trust anchor.[11] An example of a trust anchor is below in figure 3.7. This record is stored on the client, and is a copy of the parent zone's KSK.

The trust anchor is the root of the trust, and the client can use this to verify the parent zone's KSK, which then can be used to verify the ZSK, which then can be used to verify the records from the zone. This chain continues onto the child zone, where the parent can verify the child zone's KSK with the DS record, and then use that KSK to verify the ZSK, which then

```
trust-anchors {
        lab. static-key 257 3 8 "AwEAAdyubupVYO97+QR5LxX3eIh8vxOI5bxfVz6ieZqMF19e
5Ul2SD+MyyY5iM1VRutYEpLmfbMDDuW3ZElJzN4BRP8huMuzRf8V4nAuB3poxIUs+kRywr4n6et46oCBF
7oWmcw792jAKKXFiuegwZ3s5NvrZufCckq5wzgaPvMxJLZLkalddzkc0KuvwAtmGsOTT7lkn3sx59JgWG
SYiL+3FNZFSZPlcDyw3QuIO1hN+Vq2NwuSR2LkOFX3miCKx4d07hr27TtP3LGJiBhcyMeJRT3i4bOBbCt
wGelA6xq5i4k1fg2+R2e+2VZtqk35GJsh8c0yqKoahz96ZCSrWCHm690=";
};
```

Figure 3.7: An Example of my Trust Anchor

can be used to verify the records from the child zone. This is a repetitive process, where the parent can verify the child, and then the child can verify the next child, and so on. [12]

3.4 DNSSEC Options

Section to talk about my dnssec policy, key management, etc.

My client is the validator, with dnssec-validation set to yes. It has a trust anchor that was manually copied from the parent zone.

3.4.1 DNSSEC Policy

I adjusted the settings in the options configuration file, mainly to specify what algorithm to use, but also to set specifics for the keys. This is seen below in figure 3.8. Setting these settings also means that the keys are generated in the background with these specifications.

```
dnssec-policy rsa-policy {
        keys {
                ksk lifetime 365d algorithm RSASHA256 2048;
                zsk lifetime 60d algorithm RSASHA256 1024;
        };
};
```

Figure 3.8: This is what I added to my dnssec policy I created called ”rsa-policy”. This was then added to the zone configuration to implement this policy.

3.4.2 Signing

Right now, I am using Inline Signing, which automatically generates keys and signs the files. This setting also creates and keeps a separate signed zone file. Instead of running the dnssec-signzone and keygen command, the server automatically signs the zone when it starts up. This is makes resigning the zone when it expires on the server easier, and also makes it easier to manage the keys. The server will automatically generate the keys for the zone, and then sign the zone with those keys. I just specified the algorithm, size, and time to live for the keys. It also reloads the zone automatically when the keys expire or zones are changed, which is a nice feature. This is a good option for this project, because it allows me to focus on the validation part of the project, rather than the signing part.

3.5 DNSSEC In The Environment

With DNSSEC on the internet, there is normally a validating resolver separate from the client machine that authenticates and validates the answers from the server before the client gets the response back. Normally, this is done recursively. For this study, there is no recursion and instead the answers will be manually validated.

Below is an example of the Chain of Trust in the environment figure 3.9.

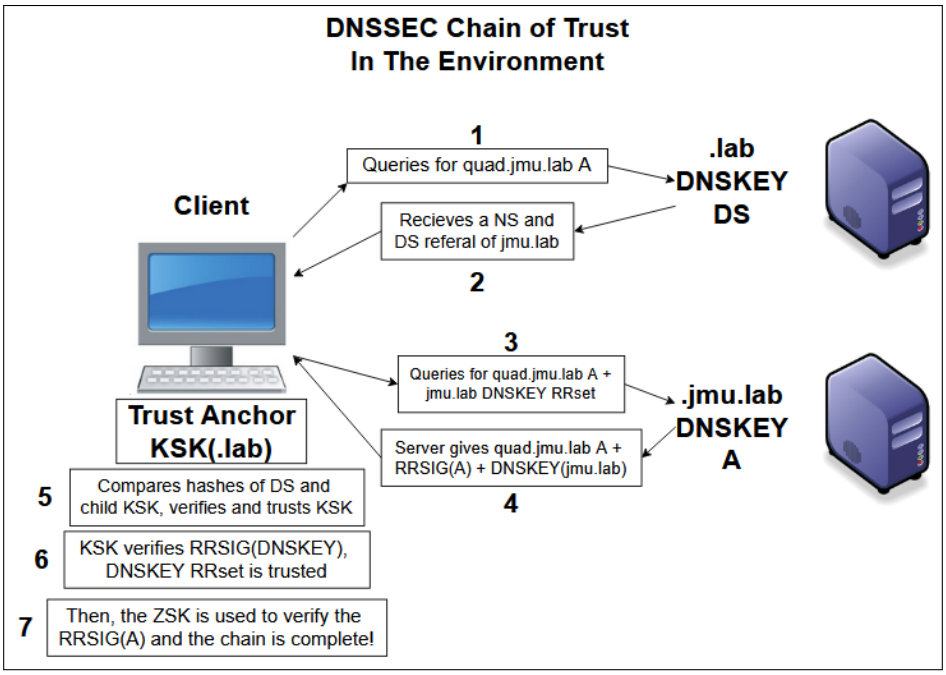

Figure 3.9: A diagram showing the Chain of Trust, with the trust anchor at the root, and the parent and child zones below. The arrows indicate the direction of trust, with each level trusting the next.

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Appendix

A.1 Procedure Manual